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Questions

Q1.

Which of these molecules can rotate the plane of plane-polarised light?

- ☐ **A** H_2NCH_2COOH
- ☐ **B** $HOCH_2CH_2COOH$
- ☐ **C** $ClCH_2C(CH_3)(Cl)COOH$
- ☐ **D** $H_2NC(CH_3)_2COOH$

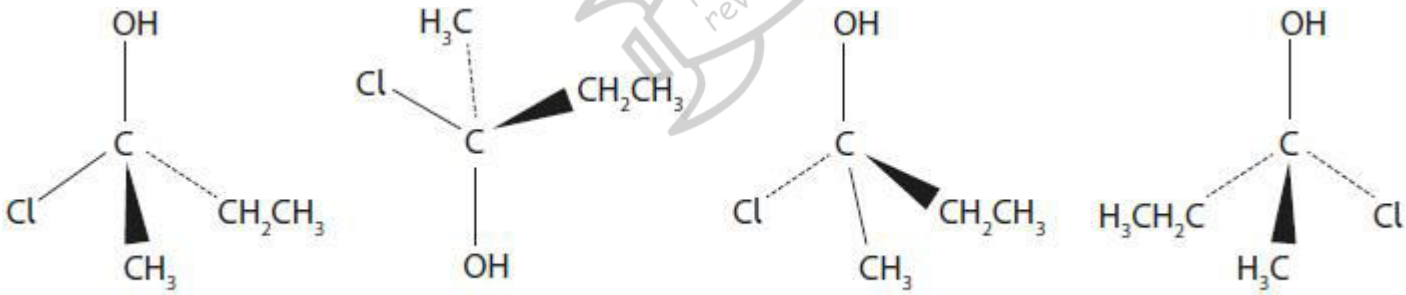
(Total for question = 1 mark)

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Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these structures is **not** identical to the others?



- ☐ **A** Structure A
- ☐ **B** Structure B
- ☐ **C** Structure C
- ☐ **D** Structure D

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(Total for question = 1 mark)

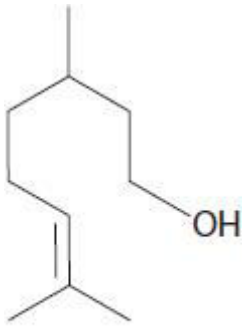
Q3.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Citronellol is found in rose and geranium oils.



citronellol

The type(s) of stereoisomerism shown by citronellol is

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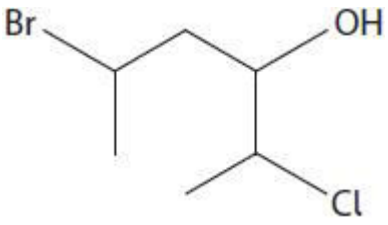
- ☒ **A** optical and geometric isomerism
- ☒ **B** optical isomerism only
- ☒ **C** geometric isomerism only
- ☒ **D** neither optical nor geometric isomerism

(Total for question = 1 mark)

Q4.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

How many optical isomers does this molecule have?



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- ☒ **A** 2
- ☒ **B** 3
- ☒ **C** 6
- ☒ **D** 8

(Total for question = 1 mark)

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Q5.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A sample of a bromoalkane, RBr, containing a single optical isomer reacts with hydroxide ions in an S_N1 mechanism.

$$RBr + OH^- \rightarrow ROH + Br^-$$

The alcohol formed is a racemic mixture.

From this information, it can be deduced that RBr is most likely to be

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☐ **A** primary only

☐ **B** secondary only

☐ **C** tertiary only

☐ **D** primary, secondary or tertiary

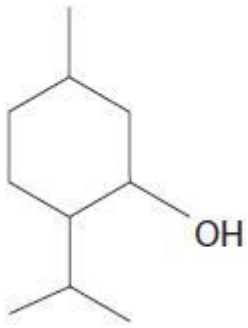
(Total for question = 1 mark)

Q6.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Menthol has a number of medicinal uses. It can be extracted from the peppermint plant.
The structure of menthol is

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How many chiral centres are there in one molecule of menthol?

☐ **A** 1
☐ **B** 2
☐ **C** 3
☐ **D** 4

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(Total for question = 1 mark)

Q7.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When a single enantiomer of 2-iodobutane reacts with sodium hydroxide, the product is a mixture of enantiomers. This is because

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- ☐ **A** the reaction is a nucleophilic substitution
- ☐ **B** the reaction proceeds via a negatively charged transition state
- ☐ **C** the reaction proceeds via a carbocation intermediate
- ☐ **D** the reaction rate depends on the concentration of 2-iodobutane

(Total for question = 1 mark)

Q8.

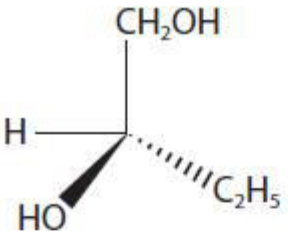
Which pair of compounds can form a racemic mixture when mixed?

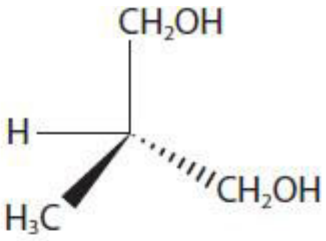
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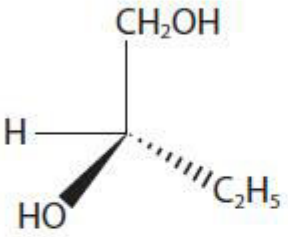
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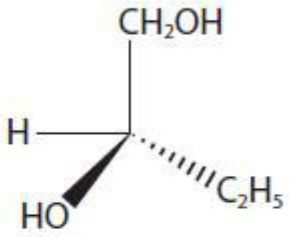
☐ **A**





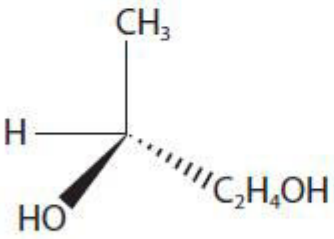
☐ **B**

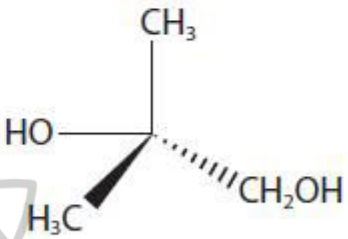




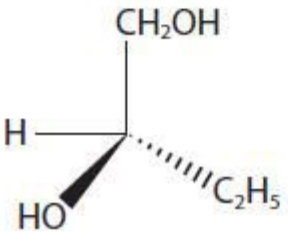
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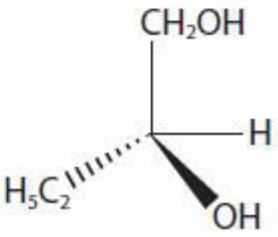
☐ **C**





☐ **D**





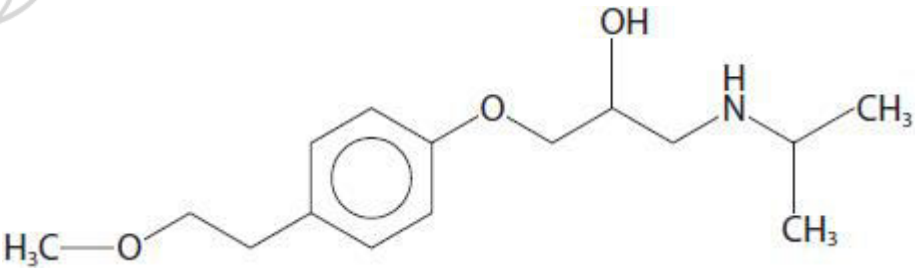
(Total for question = 1 mark)

Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Metoprolol is a drug used to treat heart problems.

The structure of metoprolol is shown.



How many chiral centres are there in a molecule of metoprolol?

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- **A** 0
- **B** 1
- **C** 2
- **D** 3

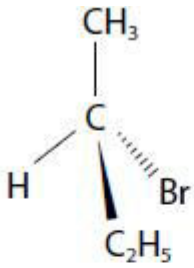
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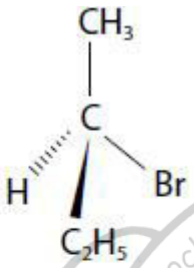
Q10.

When mixed in equimolar quantities, which pair of molecules will **not** rotate the plane of plane-polarised light?

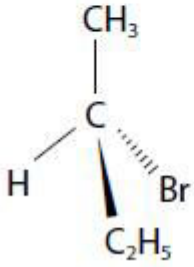
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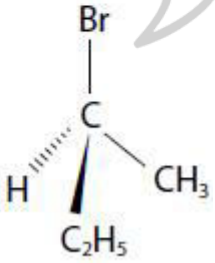
A



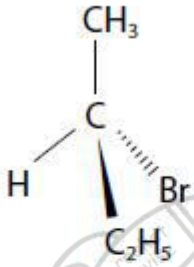


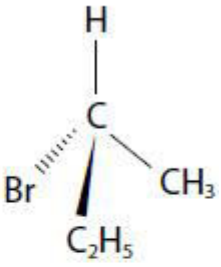
B





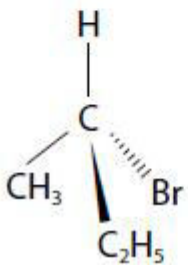
C

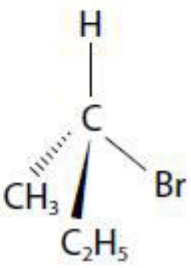




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D





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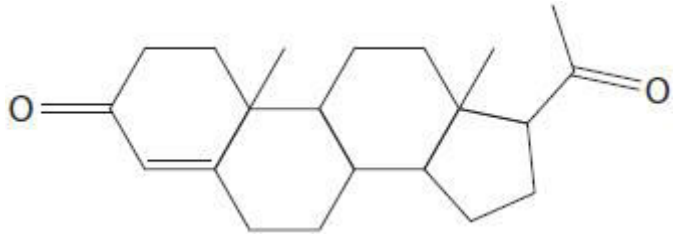
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(Total for question = 1 mark)

Q11.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

How many chiral centres are in the molecule shown?



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- ☐ **A** 4
- ☐ **B** 5
- ☐ **C** 6
- ☐ **D** 7

(Total for question = 1 mark)

Q12.

This question is about the hydrolysis of halogenoalkanes with hydroxide ions.

(a) The initial rates for the hydrolysis of 2-bromobutane, $CH_3CH_2CHBrCH_3$, were obtained for different initial concentrations of the reactants.

Both 2-bromobutane and the product formed in this reaction exist as pairs of stereoisomers.

$$CH_3CH_2CHBrCH_3 + OH^- \rightarrow CH_3CH_2CH(OH)CH_3 + Br^-$$

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Experiment	Initial concentration of 2-bromobutane / mol dm^{-3}	Initial concentration of hydroxide ions / mol dm^{-3}	Initial rate of reaction / mol dm^{-3} s^{-1}
Experiment 1	0.150	0.150	0.027
Experiment 2	0.300	0.150	0.054
Experiment 3	0.450	0.300	0.162

All rates were determined at the same temperature.

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(i) State what is meant by the term stereoisomer.

(1)

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(ii) Show that the data in the table are consistent with the S_N2 mechanism for this hydrolysis of 2-bromobutane.

(3)

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(iii) In one experiment, a single stereoisomer of 2-bromobutane was hydrolysed.
Explain the stereochemistry of the product of this reaction.

(3)

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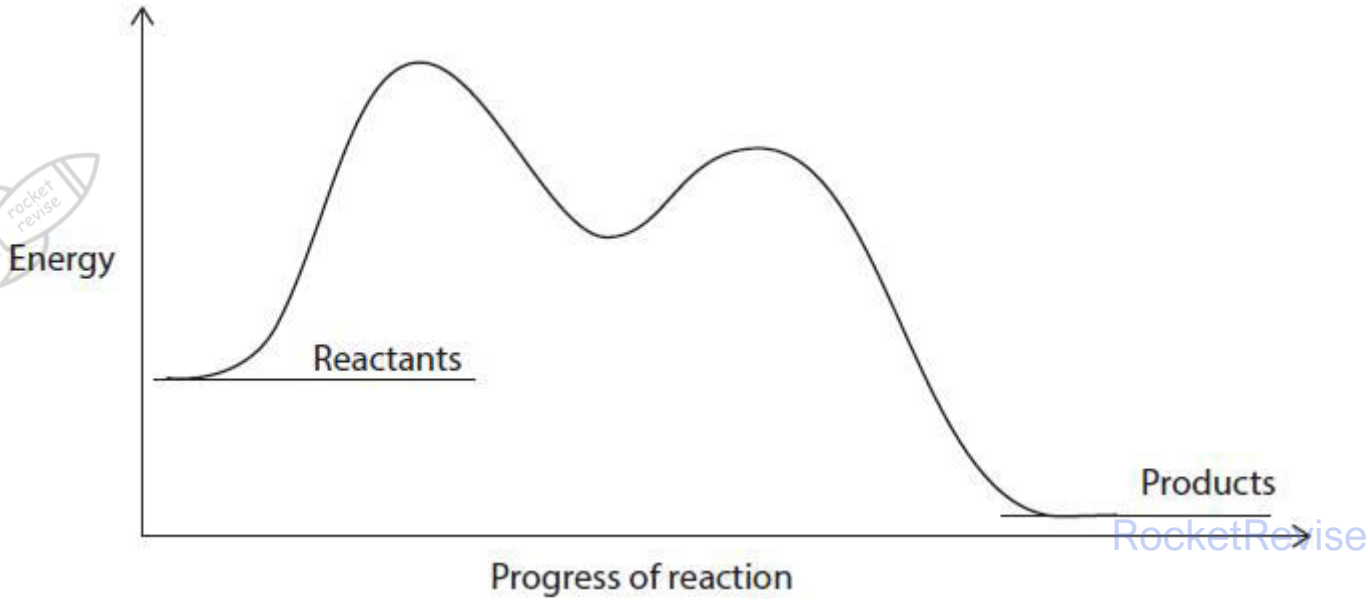
(b) 2-Bromo-2-methylbutane, $(CH_3)_2CBrCH_2CH_3$, can also be hydrolysed with hydroxide ions. The mechanism for the reaction is S_N1.

The reaction profile is shown.

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Explain how all the features of this reaction profile are consistent with an S_N1 mechanism.

(3)

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(Total for question = 10 marks)

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